

CLASS 19+20 HOMEWORK Tailwind CSS + Forms & Validation

XELI AI Labs • Frontend Engineering • Due: Next Class

5–6 hours	100 pts + drills	■ Drills: NO AI
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■ **What This Is**

Your mini project gets two upgrades tonight: it looks professional (Tailwind) and it handles user input properly (validated forms). Do the 20 quick drills first — without AI — to build muscle memory. Then use AI for the bigger restyle and form work.

Part 0: 20 Quick Tailwind Drills — Do WITHOUT AI (~30 mins)

■ **Why No AI Here**

For each item, write the Tailwind className(s) that match the description. Test each one in play.tailwindcss.com. These are short on purpose — the repetition builds the muscle memory that makes everything else faster.

1. A div with 16px padding on all sides	11. A div with a 2px solid grey border
2. A div with 24px padding top/bottom, 12px left/right	12. Small, grey-500 coloured text
3. White text on a dark blue background	13. Button padding: 16px horizontal, 8px vertical
4. Bold, centered, large text	14. A div at full width of its parent
5. Rounded corners (large) + medium shadow	15. Grid: 1 column mobile, 3 columns on lg
6. Flex container, items centered both ways	16. Extra-large heading with 16px bottom margin
7. Flex container with space between children	17. Input with an indigo focus ring
8. A 2-column grid with 16px gap	18. Disabled button: reduced opacity, no pointer
9. A button that darkens on hover	19. Auto vertical spacing (16px) between children
10. A 48x48px perfectly circular image	20. Card that lifts + grows shadow on hover, smoothly

Answers are in CLASS19-20-CODE-SAMPLES.md — check AFTER attempting, not before.

PROJECT 1 Restyle Your Mini Project with Tailwind 50 pts

■ In Plain English

Take the mini project from Class 17+18 and give it a real visual upgrade. By the end, it should look like something you could screenshot for a portfolio — and there should be ZERO custom CSS files left.

REQUIREMENTS

- Install Tailwind in your mini project (npm install tailwindcss @tailwindcss/vite)
- Restyle the Navbar: spacing, colours, active link highlighting using Tailwind
- Restyle all cards (movie/recipe/country cards): padding, rounded corners, shadow
- Responsive grid: 1 column mobile → 2 columns tablet (md:) → 3-4 columns desktop (lg:/xl:)
- Add hover states to cards (shadow increase or slight lift) and buttons (colour change)
- Restyle the detail page: clear typography hierarchy, consistent spacing
- Delete all .css files except index.css (which only contains the Tailwind import)
- Test responsiveness: resize browser through mobile, tablet, desktop widths

SETUP REMINDER

```
npm install tailwindcss @tailwindcss/vite // vite.config.js import tailwindcss from '@tailwindcss/vite';
export default defineConfig({ plugins: [react(), tailwindcss()], }); // index.css – replace everything with:
@import "tailwindcss"; // Then: restart npm run dev (full restart, not just refresh)
```

AI TIPS FOR THIS PROJECT

- Convert existing CSS: "Convert this CSS to Tailwind utility classes. Explain each class: [paste CSS]"
- Explore options: "Give me 3 Tailwind class combinations for a movie card — minimal, bold, dark mode."

BONUS POINTS

- ★ Add a dark mode toggle using Tailwind's dark: prefix
- ★ Add subtle entrance animations using Tailwind's animate- utilities

PROJECT 2 Validated Search / Contact Form 50 pts

■ In Plain English

Add a properly validated form to your mini project — either a search bar (filters your data) or a contact form (collects user info). Built with React Hook Form, styled with Tailwind, with real error messages.

REQUIREMENTS

- npm install react-hook-form
- Build a SearchForm OR ContactForm component using useForm()
- Minimum 2 fields with validation rules (e.g. required + minLength, or required + pattern)
- Each field has a custom error message (not generic "invalid")
- Conditional border colour: red when a field has an error
- Error message displayed below each invalid field
- Submit button uses isSubmitting to show a loading state
- If building a SearchForm: connect it to actually filter/search your existing data
- If building a ContactForm: console.log the submitted data (no real backend needed)

STARTER TEMPLATE

```
import { useForm } from 'react-hook-form'; const MyForm = () => { const { register, handleSubmit, formState: { errors, isSubmitting } } = useForm(); const onSubmit = async (data) => { // your logic here (search, console.log, etc.) }; const inputClass = (err) => `w-full p-2 border rounded ${err ? 'border-red-500' : 'border-gray-300'}`; return ( <form onSubmit={handleSubmit(onSubmit)} className="space-y-4"> <input className={inputClass(errors.fieldName)} {...register('fieldName', { required: 'This field is required', minLength: { value: 2, message: 'Must be at least 2 characters' } })} /> {errors.fieldName && <p className='text-red-500 text-sm'>{errors.fieldName.message}</p>} <button type='submit' disabled={isSubmitting} className='bg-indigo-600 text-white px-4 py-2 rounded disabled:opacity-50'>{isSubmitting ? 'Submitting...' : 'Submit'} </button> </form> ); };
```

AI TIPS FOR THIS PROJECT

■ Regex help: "Write a React Hook Form pattern validation for [field type]. Include an error message." Then ALWAYS follow up: "Break down this regex character by character."

■ Debugging: "My form submits even when fields are empty. Here's my code: [paste]. What's wrong?"

BONUS POINTS

- ★ Add a 3rd field with a different validation type (number range, custom regex)
- ★ Add a success message that appears after valid submission

Grading Rubric

Component	What We Look For	Pts
Drills (Part 0)	20 correct/reasonable Tailwind class answers — completed without AI	—

Tailwind Restyle	No custom CSS · responsive grid · hover states · consistent spacing	50
Validated Form	useForm() · 2+ validation rules · custom messages · error styling · isSubmitting	50

Two of the most visible upgrades a frontend project can get — done in one night. Take a screenshot when you're done; you'll want it for your portfolio later.